Unit 15 Ramping Experiment Exposure: Trading off speed.

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(1) Quality and Risk 171

When you're running an online controlled experiment (A/B test), after you've launched it to a small percent of users, you ramp up the exposure (e.g. 1% → 10% → 50% → 100%) to get faster results and broader coverage. But this ramp-up introduces risk.

- Risk is that chance that your experiment hurts your users, product or business as it scales.
- Quality is about how reliable and valid your experiments result are.

→ The key is balancing quality (data accuracy) with risk (user/business harm).

Real-World Analogy: "New Medicine Trial"

Imaging testing a new medicine.

- You first test it on 1% of patients. You monitor for side effects.
- if it looks safe, you increase to 10%. Then 50%, then the entire population.

But:

- ✓ If side effects appear early, you stop and investigate
- ✗ If you rush too fast, the drug might harm many people.

In experiments:

- The treatment variants is like the new drug.
- The users ~~users~~ are your patients
- Bad metrics (drop in revenue, engagement etc) are your side effect.

Decomposing Quality and Risk

Concept — Meaning in Experimentation.

Risk → The probability and potential impact of negative outcomes as exposure grows. Example: losing revenue, annoying users, bugs.

Quality → The statistical Power and reliability of your results (e.g. confidence intervals shrinks, variance stabilizes).

Ramping too fast → Higher risk, but faster answers
Ramping too slow → safer, but delays decisions.

(1) Mathematical Foundation

We can think of risk as:

Risk = Probability of Harm × Magnitude of Harm

Suppose:

- Probability of revenue drop = 5%
- Potential revenue loss if happens = $ 100,000

Expected Risk = 0.05 × 100,000 = $ 5,000.

So even if chance is small, if impact is big, you might want to slow the ramp.

(2) Summary

Key Point	Explanation.
Ramping = increasing exposure	→ Speeds up testing but introduces risk.
Quality = Trustworthy results	→ Need good instrumentation, design and power.

Risk = Probability x Harm → Even small % can cause large impact at scale.

Goal → Trade off speed, quality, and risk smartly using metrics and strategy.

② Related Concepts (Future linked Topics).

* "SQR Ramping Framework" (next section in Unit 18)
  - SQR = Speed, Quality, Risk
  - The trade off: faster ramps vs. risk exposure vs. experimental quality.
  - Decisions about ramp speed are based on balancing these three.

* "User-Level Randomization" (Unit 14)
  - Quality depends on proper randomization at the user level.
  - Bad randomization → biased results → false safety signals → risky decisions during ramp.

* "Culture of Instrumentation" (Unit 13)
  - If you don't measure well (bad instrumentation), you won't detect risk.
  - Strong telemetry = catch harmful ramps early.

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② What is Ramping? (page 171)

① What is Ramping?

→ Ramping is the process of gradually increasing the exposure of an experiment (variant or treatment) to more users over time.
Instead of launching a new feature to 100% of users immediately, you start small (e.g. 1%), monitor metrics, and then "ramp" up to more users in stages: 1% → 5% → 10% → 25% → ... → 100%.

Why do we ramp?
To balance three competing goals:
1. Speed - get results faster.
2. Quality - ensure statistical reliability.
3. Risk - minimize potential harm to user/business.

This directly leads into the SQR framework (Speed, Quality, Risk) which comes in the next sections.

Real-world Analogy: "New Restaurant Dish Rollout"

Imagine you're a chef introducing a new dish!
- You first give it to 1 table.
- If they like it, you try it on 5 tables.
- Then 10, then 50 ... and finally put it on the main menu

Why not go all-in at once?
Because:
- The dish might be too spicy, cause food poisoning, or just be a bad experience.
- You want to test it small, learn, adjust, then scale safely

Ramping is like soft launching a change to reduce business/user damage, learn faster and be cautious.

How Ramping works in A/B Tests.

A simplified process:

1. Start small: e.g. 1% users see the treatment.
2. Monitor metrics: revenue, error rates, engagement, etc.
3. If safe, increase exposure: to 5%, 10%, 25% etc.
4. Eventually reach 100% (if successful).

Each step is called a "ramp", and the full sequence is the ramping process.

(2) Mathematical Concept Behind Ramping

#1. Expected Harm Function.

If a variant has a negative effect, ramping too fast causes longer cumulative damage.

If the variant causes a -2% revenue impact per user, and we expose 1 million users, the cumulative loss increases as we ramp.

We model it as:

$$\text{Cumulative Harm} = \sum_{i=1}^{n} \text{Exposure}_i \times \text{Impact}_i$$

where:

- Exposure_i: number of users in ramp stage i
- Impact_i: negative metric change in that change.

#2. Statistical Power Improves with Ramp

Larger sample → smaller standard error → better CI

So:

- Ramp fast = faster answers, more precision
- Ramp slow = safer, but slower decision-making.

③ Related Concepts

Concept	Connections
Quality & Risk (previous section)	→ Ramping is how you control risk while still collecting quality data.
SQR Ramping framework (next)	→ Ramping is the foundation of how you balance Speed, Quality, Risk.
User-level Randomization (Unit 14)	→ Ensures correct exposure calculation and consistent ramping
Instrumentation (Unit 13)	→ You need good telemetry to detect metric issues during each ramp stage.

④ Summary

Concept	Meaning.
Ramping	Gradually increasing % of users exposed to an experiment variant.
Goal	Learn safely, move quickly, reduce harm.
Risk	Higher when you ramp faster if variant is flawed.
Real Example	like launching a new product feature to increasing audiences.
Math Basis	Harm grows with exposure, but statistical power improves too.

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(4) SQR Ramping Framework (page 172).

SQR Ramping Framework (page 172).

(1) What is the SQR Ramping Framework?

SQR = Speed, Quality, Risk

This framework helps you decide how fast to ramp an experiment by trading off:

Component	Meaning
S = Speed	How quickly you want to get results and ship features.
Q = Quality	Statistical reliability and confidence in your experiments.
R = Risk	Probability and magnitude of harm to users or business.

Goal: Make smart, evidence-based ramping decisions by considering:

- How quickly you want to reach a conclusion (Speed).
- How trustworthy your results are (Quality).
- How much damage could occur if you're wrong (Risk).

Real-world Analogy: "Product Recall vs Product Launch"

Imagine you're launching a new mobile phone:

- Speed: You want to sell units quickly to beat competitors
- Quality: You need good reviews and reliable testing (battery, screen, camera)
- Risk: A faulty battery could explode - massive brand damage.

You can't ignore quality or move too fast without checking for defects.

You balance: "how fast we move" vs. "how safe it is" vs "how sure we are it's good".

How S, Q, and R Interact.

This framework acts like a triangle:

- If you increase speed, you might:
  - lower quality (less data)
  - increase risk (more users harmed)
- If you optimize for quality (e.g. wait for long p-values)
  - Ramp is slower
  - safer, but delays decisions.
- If you minimize risk (slow, careful ramps):
  - It slows down speed
  - Quality may improve, but at high time cost

You must balance all three, not optimize just one.

(2) Math Model Behind SQR

Let's define some abstract relationship

1. Risk at ramp i:

$$R_i = P_i \times H_i$$

where

- P_i: probability of negative effect.
- H_i: magnitude of harm at exposure level i

2. Statistical Quality increases with more users:

$$\text{Standard Error (SE)} \propto \frac{1}{\sqrt{n}}$$

3. Speed = how quickly we reach significant conclusion
   - Increases with longer ramps
   - Decrease with over-caution

(3) Interlinked Topics

Topic	How it Connects
"What is Ramping?"	→ SQR framework provides the why and how behind ramping stages.
"Quality & Risk"	→ Two pillars of SQR : the need to balance metric accuracy with potential user harm.
"Four Ramp Phases"	→ SQR helps decide when to move from one ramp phase to the next
Unit 14 - User_level Randomization	→ Quality depends on proper randomization : Flawed randomization breaks SQR reliability.
Unit 13 - Instrumentation	→ Without good telemetry, you can't measure Q or R correctly, invalidating the SQR tradeoff.

(4) Summary

SQR Component	Meaning	Role in Ramping
Speed	How fast you ramp and decide.	Push to deliver value quickly.
Quality	Statistical reliability	Ensures you don't expose users to bad changes
Risk	Business or user damage potential	Avoid harming real users as ramp increases.

⇒ Ramping decisions should be based on a measured balance of these three - not gut feeling.

(4) Four Ramp Phases 173

1) What are the Four Ramp Phases?

The authors define four structured phases of ramping an online experiment. These help balance speed, quality, and SQR framework and ensure safe, confident decision-making at every stage.

The Four Ramp Phases Are:

Phase	Exposure	Purpose
1. Pre-Launch Check (smoke test)	~ 0.1% - 1%	Test plumbing, instrumentation, and basic functionality
2. Safe Ramp-up	1% → 5% → 10%	Catch large bugs/issues with minimal risk.
3. Data Ramp	10% → 25% → 50%	Collect statistically meaning data for analysis
4. Final Ramp (GO/NO-GO)	50% → 100%	Operational rollout for all users if results are safe.

1. Pre-Launch check (Smoke Test)

- Goal: Validate that your experiment is wired correctly - randomization, metrics, event tracking etc.
- Exposure: Very small (0.1 - 1%)
- Checks
  - ↳ logging works
  - ↳ Users assigned correctly to treatment/control
  - ↳ No crashing or UI failures.

- *Analogy: Like turning on a new car engine for 5 seconds to see if it starts - before driving it.

2. Safe Ramp-up.

- Goal: Detect high-risk problems (e.g. major performance or revenue drops) before too many users are impacted.
- Exposure: Gradually increase (e.g. 1% → 5% → 10%)
- Action: Monitor key metrics (engagement, errors, conversions)

* Analogy: Letting your new product be used by a small test market before a national launch.

| Risk is low, but enough users see the features to match "deal-breakers" issues.

3. Data Ramp

- Goal: Collect enough data for statistical significance
- Exposure: Mid to Large (e.g. 10% → 25% → 50%.)
- checks
  - ↳ use p-values, confidence intervals, CUPED adjustments
  - ↳ Evaluate: GO/NO-GO: is it safe to move forward?

* Analogy: Running a clinical trial on hundreds of patients — you're validating effectiveness and safety with real data.

4. Final Ramp (Go/No-Go Decision)

- Goal: If experiment is successful, deploy to 100% of users
- Exposure: 50% → 100%
- checks
  - still monitor key metrics
  - Plan rollback if sudden regression appears.

* Analogy: Full product launch with rollback switch on standby in case something goes wrong.

2) Mathematical framing Behind Phases.

Risk Vs. Exposure Curve.

~~Let's model risk~~

Let's model risk exposure across phases.

Let:

- R_i = Risk at phase i = $P_i \times U_i \times L_i$
  - P_i = probability of failure
  - U_i = number of users exposed.
  - L_i : average loss per user.

(2) How This Ties Into Other Concepts.

Topic	Connection
SQR Framework	These phases are practical steps to balance SQR (Speed, Quality, Risk).
What is Ramping	The four phases provide the structured blueprint for safe ramping.
Quality and Risk	Safe ramp + data ramp let you minimize risk while increasing data quality.
Post-Final Ramp	Comes after 100% launch - continued monitoring and learning happen here.
User-level-Randomization (Unit 14)	Must ensure correct exposure logic throughout all phases, or else all results are invalid.
Culture of Instrumentation (Unit 13)	Strong telemetry is critical to detect problems early, especially in phase 1 and 2.

(4) Summary Table

Phase	Exposure	Main Goal	Risk	Speed	Quality
Smoke Test	~1%	check wiring & instrumentation	Low	Slow	Low
Safe Ramp	1% - 10%	Catch	Low	Medium	Medium
Data Ramp	10% - 50%	statistical confidence	Medium	Medium	High
Final Ramp	50% - 100%	full rollout	High	Fast	High

⑨ Post Final Ramp 176.

① What is "Post final Ramp"?

→ The "Post final Ramp" phase comes after your experiment has been ramped to 100% of the user base.

> You've made the Go decision → Now all users are exposed to the treatment.

But the experiment doesn't end just because = 100%
This phase focuses on:

Key focus	Why it Matters.
Monitoring for regression	Bugs or side effect may appear later.
Long-term impact tracking	short-term lift may vanish or reverse
Operational readiness	Do logging, telemetry, and rollback systems work under 100% load?
Metric drift	Behavior or data quality might change over time.

Real-World Analogy: "New Bridge Opens to Public".

Imagine you build a new bridge

- Phase 1: Test small cars.
- Phase 2: Test buses
- phase 3: let as % of traffic on.
- final ramp: Open to all vehicles.

But even after full launch, you don't walk away. You

- Keep sensors on to detect stress/load
- Monitor if vibrations increase
- Track if accidents increase
- Evaluate long-term reliability

That is Post final Ramp in online experiments.

② Statistical Foundation

1. Regression to the Mean
   - A strong treatment effect observed early may fade over time
   - Post Final Ramp lets you detect this natural drop

2. Delayed Impact Detection
   - some effects (e.g. churn, loyalty, lifetime value) are long-lagged metrics.
   - You need long-term observation beyond the ramp.

③ Links to other Units

Related Concepts	Overlap
Four Ramp Phases	→ Final ramp is NOT the end - Post Final Ramp ensures long-term health.
Quality & Risk	→ Risk may not show up immediately - long-term quality assurance depends on this phase.
Culture of Instrumentation (unit 13)	→ You need strong logging, metric alerts, anomaly detection after rollouts.
Twyman's Law (Unit 3)	→ Be skeptical of unexpected good results - they might regress post-launch.
Segment Heterogeneity (unit 9)	→ Post Final ramp can reveal sub group-level differences missed in aggregate data analysis.

(4) Summary

Aspect	Description
Definition	Phase after full exposure to all users
Purpose	Monitor for long-term issues, confirm robustness
Key Risks	Metric decay, regressions, missed bugs
Statistical Concept	Regression to mean, delayed effect detection
Tools Needed	Alerting, anomaly detection, long-term logging
Time Window	Often 1-4+ weeks after full rollout, depending on metric type.

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